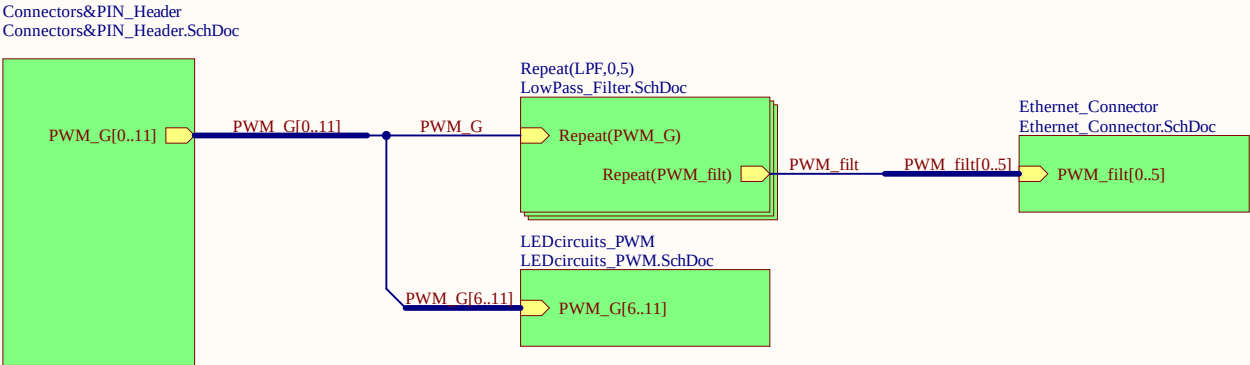




LOGO1

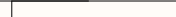


Rev02
Revision1

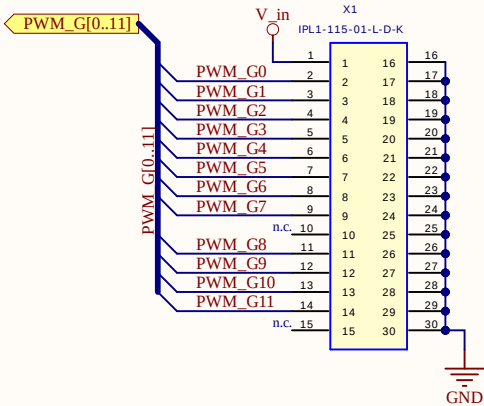
Serial
Serial1

uz_per_loopback_tutorial
Project1

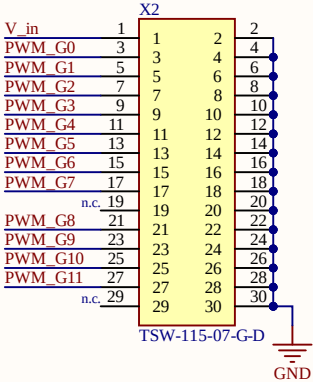
J. Hahn
Designer1

Title Interface_TopSheet.SchDoc		UltraZohm www.ultrazohm.com Date: 02.04.2026 Sheet 1 of 10	
Revision: Rev02	Design Engineer: J. Hahn		
Project: UZ_PER_loopback_tutorial.PrjPcb			

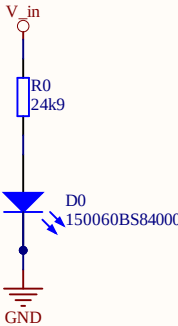
Input connector



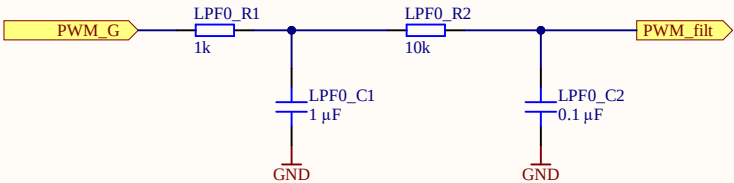
Pin header



LED for input voltage - Blue

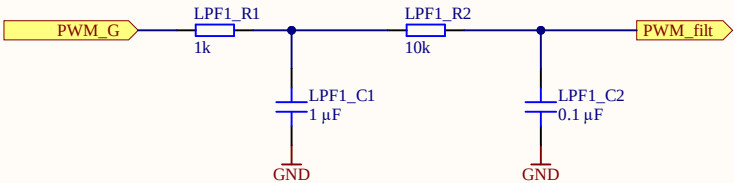


Low pass filters



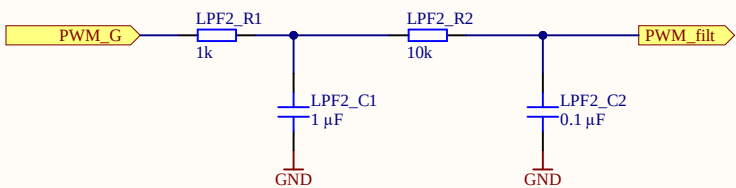
▲ Low pass filter
cutoff-frequency
 $f_c =$
 $1/(2 \cdot \pi \cdot \sqrt{R1 \cdot C1 \cdot R2 \cdot C2}) =$
159 Hz

Low pass filters



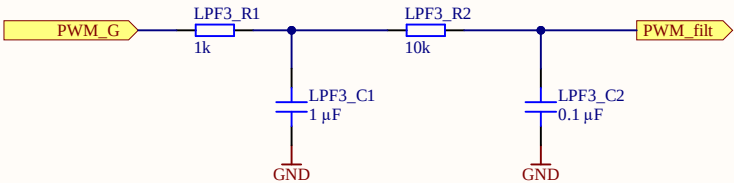
Δ Low pass filter
cutoff-frequency
 $f_c =$
 $1/(2\pi \sqrt{R1 \cdot C1 \cdot R2 \cdot C2}) =$
159 Hz

Low pass filters



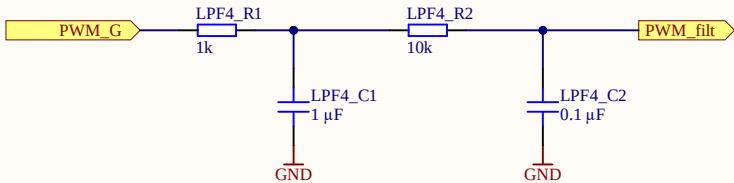
Δ Low pass filter
cutoff-frequency
 $f_c =$
 $1/(2 \cdot \pi \cdot \sqrt{R1 \cdot C1 \cdot R2 \cdot C2}) =$
159 Hz

Low pass filters



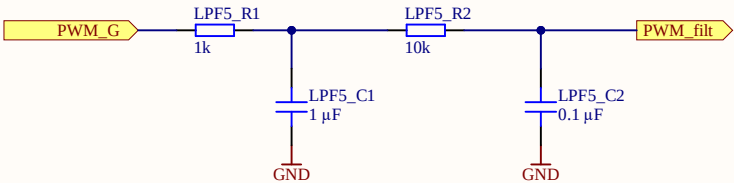
Δ Low pass filter
cutoff-frequency
 $f_c =$
 $1/(2 \cdot \pi \cdot \sqrt{R1 \cdot C1 \cdot R2 \cdot C2}) =$
159 Hz

Low pass filters



▲ Low pass filter
cutoff-frequency
 $f_c =$
 $1/(2 \cdot \pi \cdot \sqrt{R1 \cdot C1 \cdot R2 \cdot C2}) =$
159 Hz

Low pass filters



Δ Low pass filter
cutoff-frequency
 $f_c = 1/(2 \cdot \pi \cdot \sqrt{R1 \cdot C1 \cdot R2 \cdot C2}) = 159 \text{ Hz}$

A

A

B

B

C

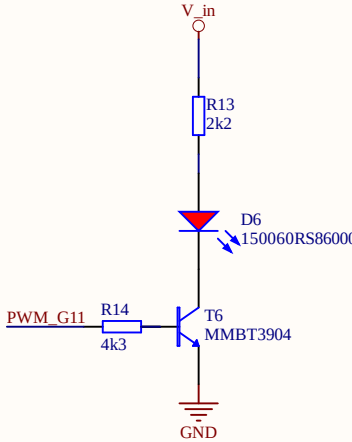
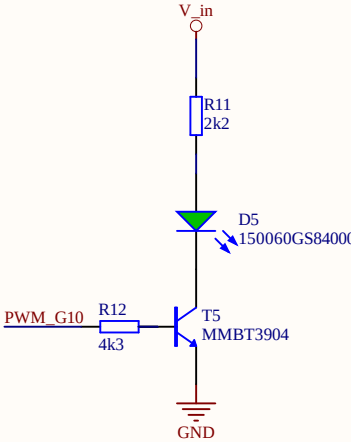
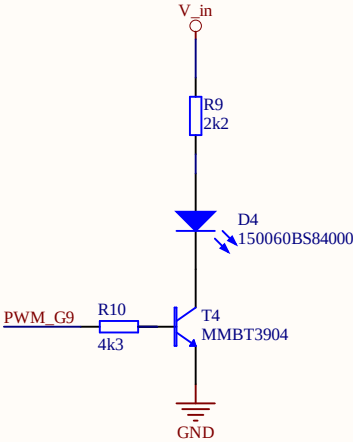
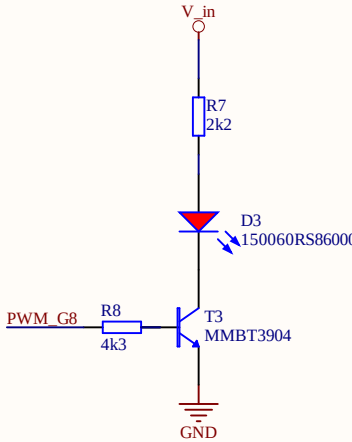
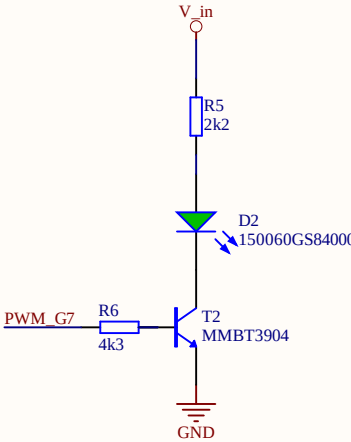
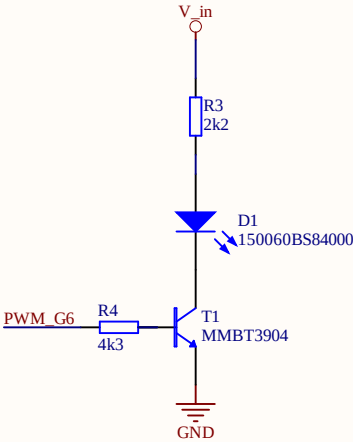
C

D

D

PWM_G[6..11]

PWM_G6
PWM_G7
PWM_G8
PWM_G9
PWM_G10
PWM_G11



Title LEDcircuits_PWM.SchDoc

Revision: Rev02

Design Engineer: J. Hahn

Project: UZ_PER_loopback_tutorial.PrjPcb

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Date: 02.04.2026

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Ethernet Connectors

